Tutorial 5 - SPM (wk 10 done)

Question 1
Team 3

Question 1
The risk projection shows the following information:
60 reusable software components were planned. 42 components can be reused, 18 new components would have to be developed from scratch and new requirements.
Risk identified: 18 components have to be custom developed based on new requirements. The average component is 100 lines of code (LOC) and the cost for each LOC is RM14.00.

Risk probability: 50%

Calculate the following values using a Halstead's risk exposure metric (probability X cost):
- Cost to develop the 18 software components.
- Risk exposure

- Planned reusable components = 60
- Actually reusable components = 42
- New components to be developed = 18
- Average LOC per component = 100
- Cost per LOC = RM14
- Risk probability = 50%

18 x 100 = 1800 LOC
Cost = 1800 x RM14
= RM25200

Risk Esposure = Probability X Cost
= 0.5 X RM25200
= RM12600

Team 1

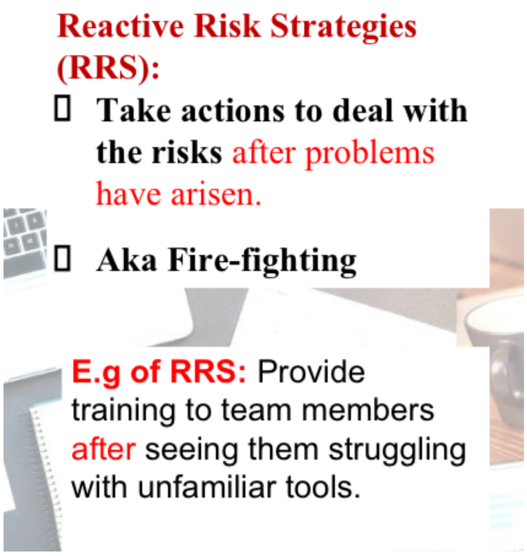
The cost to develop the 18 components
= 18 X 100 X 14
= RM 25,200

Risk exposure : RE = P X C
= 0.50 x 25,200
= RM 12,600

Question 2
Team 4

Question 2

Project risks can be dealt with by using reactive risk strategy or proactive risk strategy. Explain *reactive risk strategy* then give an example to highlight why it is a bad strategy.

Strategy type	**Reactive risk strategy**
When to take action  	**Take actions to deal with the <u>risk after problems have arisen</u>** **For when the risk is urgent to be overcome.**
Examples	**A project team manager introduces a new IDE to his software developers. Due to not familiar to the IDE, they have trouble to develop using the newly introduced IDE. The project manager sees them and instantly do research, and hire a trainer to train the team to use it.**

when comparison question come out:

5.1 Introduction

Reactive Risk Strategies (RRS):

- **Take actions to deal with the risks** after problems have arisen.
- **Aka Fire-fighting**

E.g of RRS: Provide training to team members after seeing them struggling with unfamiliar tools.

Proactive Risk Strategies (PRS):

✔ **Identify potential risks, assess the probability and impact** before any technical work begins.

✔ **Develop a contingency plan to respond to the risk** in a controlled and effective manner.
(this means preparing in advance what to do if any identified risk materialises)

E.g of PRS: Provide training to team members on unfamiliar tools before project starts.

Question 3

Consider the following potential risks in table 1

Table 1: Project risk, category, probability and impact

No.	Risks	Category	Probability of Occurrence (0-1)	**Impact* on the project (1-4)	Risk Score
1	Unstable database	TE	0.5	3	
2	Inadequate software architecture	PR	0.3	2	
3	Staff turnover in project team	ST	0.6	4	
4	Resistance to use system	CU	0.1	2	

*Impact values: 1 – negligible 2 – marginal 3 – critical 4 – catastrophic

a) Calculate the risk score for each risk listed in table 1, *rank the risks* according to their priority *using risk no* (i.e., from highest to lowest risk score). Show the workings in your calculation.

b) Identify ONE (1) risk with the **highest** risk score in part a). Propose a suitable *risk mitigation* and *monitoring actions* for the identified risk. Explain your answer.

Team1

No.	Risks	Category	Probability of Occurrence (0-1)	**Impact* on the project (1-4)	Risk Score
1	Unstable database	TE	0.5	3	0.5x3=1.5
2	Inadequate software architecture	PR	0.3	2	0.3x2=0.6
3	Staff turnover in project team	ST	0.6	4	0.6x4=2.4
4	Resistance to use system	CU	0.1	2	0.1x2=0.2

a) Risk Ranking:

1. Staff turnover in project team = 2.4
2. Unstable database = 1.5
3. Inadequate software architecture = 0.6
4. Resistance to use system = 0.2

b) Highest risk: Staff turnover in project team.

Mitigation:

- Boost Morale: Recognize good work publicly and ensure manageable workloads to prevent burnout.
- Cross-Train Team: Ensure at least two people can handle every critical task to reduce knowledge loss.
- Document Everything: Maintain up-to-date project documentation so new members can get up to speed quickly.

Monitoring:

- Conduct Stay Interviews: Regularly ask key members what keeps them happy and what might make them leave.
- Track Engagement: Watch for warning signs like increased sick days, dropped productivity, or less participation in meetings.
- Monitor Turnover Rate: Calculate the team's turnover rate montly to see if it's getting better or worse.

Team 3

Question 3

Consider the following potential risks in table 1

Table 1: Project risk, category, probability and impact

No.	Risks	Category	Probability of Occurrence (0-1)	**Impact* on the project (1-4)	Risk Score	
1	Unstable database	TE	0.5	3	0.5x3	= 1.5
2	Inadequate software architecture	PR	0.3	2	0.3x2	= 0.6
3	Staff turnover in project team	ST	0.6	4	0.6x4	= 2.4
4	Resistance to use system	CU	0.1	2	0.1x2	= 0.2

*Impact values: 1 – negligible 2 – marginal 3 – critical 4 – catastrophic

a) Calculate the risk score for each risk listed in table 1, *rank the risks* according to their priority *using risk no* (i.e., from highest to lowest risk score). Show the workings in your calculation.

b) Identify ONE (1) risk with the **highest** risk score in part a). Propose a suitable *risk mitigation* and *monitoring actions* for the identified risk. Explain your answer.

a)

1. Staff turnover in project team = 2.4
2. Unstable database = 1.5
3. Inadequate software architecture = 0.6
4. Resistance to use system = 0.2

b)

b.

- Highest Risk: Staff turnover in project team (Score = 2.4)
- Category: Staff size and experience (ST)
- Risk Mitigation Actions:
  - Implement knowledge sharing practices (documentation, code reviews, pair programming).
  - Introduce cross-training so multiple team members can cover each role.
  - Offer incentives (bonuses, career growth, recognition) to reduce turnover likelihood.
- Monitoring Actions:
  - Regularly track employee satisfaction and engagement.
  - Conduct exit interviews to detect trends in resignations.
  - Maintain a resource buffer by preparing backup staff or contractors.